WHAT HIDES MUSICAL GEOMETRY IN AUDIO REPRESENTATIONS?

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ABSTRACT

Learned representations are often evaluated by how well their attributes can be decoded. However, linear decodability does not imply that a representation's ambient geometry reflects the relationships among those attributes. Music makes this distinction directly testable because relationships between musical classes, such as keys and pitches, are explicitly defined. We examine this gap through linear decoding, class-centroid geometry, and linear-readout geometry. A log-CQT decodes key more accurately than MERT layer 12 under track-grouped cross-validation (macro F1 .392 vs. .371), yet shows no circle-of-fifths structure in its class centroids while the same structure is strong in its readout weights. We further identify two factors that determine geometric visibility. First, increasing the number of examples per centroid changes fifths geometry by a factor of 13.4, compared with only 1.7 for a thousandfold increase in pooling duration. Second, octave equivalence differs across metrics: it can be absent from global distances, weakly preserved in geodesic geometry, and substantially strengthened by a supervised rank-4 projection fitted only on GiantSteps keys, in subspaces accounting for less than 0.3% of the variance. These results show that decodability, ambient geometry, and readout geometry are distinct properties of a representation, and that musical relations can differ substantially in where and how strongly they appear.

Index Terms— music representation learning, probing, representation geometry, key detection, octave equivalence

1. INTRODUCTION

Self-supervised music audio encoders support linear probes [1] for key, pitch, genre, and instrument [2, 3], and this is often interpreted as evidence that these attributes are *represented* in the embedding. Music provides a useful setting for testing this interpretation because music theory specifies not only which classes exist, but also how they are related. Keys a perfect fifth apart share six of seven scale degrees and are adjacent on the circle of fifths, while pitches an octave apart belong to the same pitch class.

We compare what can be decoded from a representation, how musical classes are arranged by distances in the original representation space, and how those classes are arranged in the weights of a linear classifier. Across these measurements, linear accessibility does not necessarily correspond to structure in the original representation geometry. Our results trace this mismatch to two distinct sources. For clip-level key geometry, the dominant factor is the number of examples used to estimate each class centroid rather than the duration of each example. For note-level octave equivalence, whether the structure is present in the ambient metric at all depends on the encoder, and increasing the number of examples does not change that, whereas a learned projection strengthens it under a different metric. Together, these results show that the absence of structure in the original geometry does not by itself imply the absence of that structure from the representation.

2. RELATED WORK

Linear probes are the standard instrument for asking what a representation encodes [1], and music audio encoders are routinely evaluated this way, with MERT [2] and MARBLE [3] reporting probe accuracy for key, pitch, genre and instrument. Hewitt and Liang [4] argue that probe performance should be interpreted relative to control tasks designed to measure probe selectivity.

Music cognition has established well-defined relational structures for both key and pitch. Perceived key relationships follow the circle of fifths and Krumhansl–Kessler profiles [5, 6], while pitch perception separates pitch height from chroma [7]. Chroma features [8, 9] encode octave equivalence by construction and therefore provide a positive control for these relationships. Tonality-aware objectives such as STONE [10] explicitly impose key equivariance during training. Pilataki et al. [11] report a related observation in a JEPA-based transcription model, where NSynth embeddings show strong separation by timbre and pitch height but weak separation by pitch class.

Code, pre-registrations and all run artifacts: `https://github.com/Junst/topo-music`. Every threshold and kill criterion reported here was registered before the corresponding run.

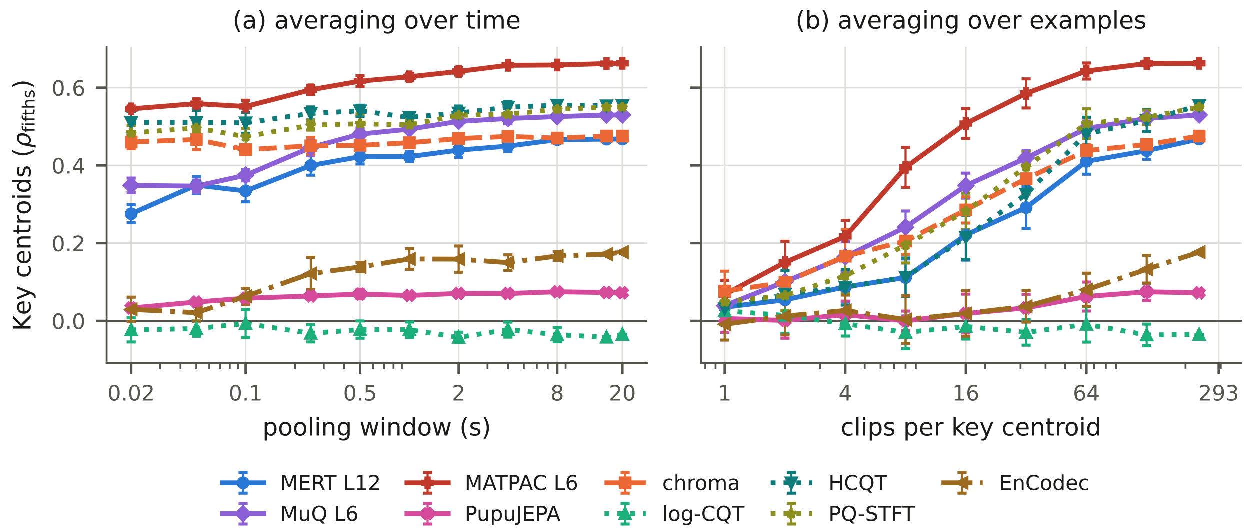

Fig. 1. Clip-level fifths geometry is affected much more by the number of clips per centroid than by temporal context. (a) Pooling-window sweep with the number of clips fixed. (b) Clips-per-centroid sweep at full-clip pooling. Error bars are one standard deviation over 5 and 10 draws.

3. METHOD

Representation decomposition. We write an item of class y as

$$z = s_y + n_y + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid y] = 0, \tag{1}$$

where s_y represents the class-dependent signal whose relations encode the structure of interest, such as key relationships or octave equivalence, n_y captures other variation that depends on the class, and ε captures variation whose conditional mean is zero within each class. Averaging examples from the same class reduces ε but not n_y, whereas a change of metric can alter how s_y and n_y contribute to the measured geometry.

Representations. Three self-supervised music encoders, MERT-v1-330M [2], MuQ-large [12] and MATPAC++ [13] in its music variant, sampled at matched fractions of depth, since the latter two are half as deep; PupuJEPA [14], a music-domain JEPA over mel patches, read from its teacher encoder; a neural codec, EnCodec 32 kHz [15] as used by MusicGen [16]; and three log-frequency front ends, a log-CQT [8, 9] at 12 bins per octave and, at 60 bins per octave, an HCQT stacked over six harmonics and a multi-scale pitch-quantized STFT. A CQT chromagram [9] folds octaves into 12 pitch classes and serves as a positive control. Frame-level representations are averaged over time to one vector per clip.

Key-centroid geometry. On GiantSteps [17], with 7035 clips covering all 24 keys, we average the vectors of tracks sharing a key into 24 class centroids and rank- correlate their pairwise Euclidean distances against three references: circle- of-fifths distance, the Krumhansl–Kessler key-profile distance, and semitone distance. Significance uses a 2000-draw permutation null over key labels, and a partial Spearman additionally removes mel-spectral centroid distance. Semitone distance is included as a control to distinguish circle-of-fifths structure from proximity in chromatic pitch space.

Readout geometry. The same probe, a multinomial logistic regression, yields 24 class weight vectors, and we apply the identical measurement to their pairwise cosine distances. All key probes use StratifiedGroupKFold with track identity as the grouping variable, since the 7035 clips come from 1763 tracks and excerpts of one track must not appear in both training and test folds.

Octave equivalence. On NSynth [18] we use real recordings only, since pitch shifting by signal processing would build the answer into the stimulus. For each instrument we fix an anchor set of pitches supporting every transposition up to 25 semitones, held constant across transpositions, and measure the mean cosine similarity between a note and its transposition by k semitones. Three statistics are reported together, as registered in advance: a local octave contrast against the mean of 11 and 13 semitones; a strict contrast against the largest similarity from 8 to 11; and the coefficient on a twelve-semitone periodic term in a fitted curve. Confidence intervals are bootstrapped over *instruments*. We require the three to agree in sign, since a positive periodic coefficient with no local peak at 12 semitones is a curve-fitting artifact.

Metrics. The same octave statistics are computed under three metrics on the same recordings. The native metric is cosine distance in the representation. The geodesic metric is the shortest path on a symmetric k-nearest-neighbor graph over the embeddings with cosine distance on the edges, which involves no supervision and no linear map, and we sweep k from 5 to 50 and report unreachable pairs rather than imputing them. The third is the supervised linear metric induced by the projection below. For the distance-based metrics we negate

Table 1. Key information, ambient geometry and readout geometry come apart, on the same 7035 GiantSteps clips and 24 keys. F1 is track-grouped macro F1 over the 24 imbalanced keys; accuracy is reported in the repository. ρ_5^{cent} is the circle-of-fifths correlation among the 24 key centroids and ρ_5^W the same among the probe's class weight vectors, while ρ_{chr} measures correlation with semitone distance and controls for proximity in chromatic pitch space. Orthogonal key codes reach a macro F1 of .613 with classes equidistant by construction and still show nothing, showing that fifths structure in ρ_5^W is not a consequence of classifier accuracy alone. In each of the first three columns blue marks the highest value among the real representations and red the lowest.

representation	F1	ρ_5^{cent}	ρ_5^W	ρ_{chr}
Controls				
analytic fifths	0.181	+0.977	+0.986	−0.085
random features	0.039	+0.009	+0.033	−0.001
orthogonal key codes	0.613	−0.008	−0.005	−0.001
Spectral front ends				
log-CQT, 12 bin/oct	0.392	**−0.035**	+0.560	+0.174
HCQT, 60 bin/oct	0.389	+0.554	+0.134	−0.038
PQ-STFT, 60 bin/oct	0.391	+0.550	+0.172	+0.112
chromagram	0.346	+0.475	**+0.692**	+0.140
Neural codec				
EnCodec 32 kHz	0.332	+0.177	+0.648	+0.041
MERT-v1-330M				
layer 4	0.379	+0.445	+0.412	−0.025
layer 12	0.371	+0.468	+0.428	+0.024
layer 24	0.340	+0.294	+0.484	+0.049
MuQ-large				
layer 2	**0.418**	+0.533	**+0.103**	+0.021
layer 12	0.351	+0.372	+0.464	+0.056
MATPAC++ music				
layer 6	0.390	**+0.663**	+0.366	−0.061
layer 12	0.329	+0.548	+0.656	−0.052
PupuJEPA				
teacher encoder	**0.198**	+0.072	+0.570	−0.007

distances before computing the same contrasts, so a positive value means the same thing throughout. **Subspaces.** Each subspace is an orthonormal basis of the top d shrinkage-LDA discriminant directions for a stated target, with d at most 11, the rank of any linear map to a twelve-class target. We report the fraction of total variance it holds and, as a control, matched random orthonormal subspaces, since projection changes the metric whatever the directions are.

4. RESULTS

4.1. Information, metric and readout dissociate

Table 1 measures all three on the same clips and keys, and the control block sets the scale. The analytic fifths and random-feature controls verify the expected positive and null regimes. Orthogonal per-key codes are highly decodable with tonal relations absent by construction. They reach a macro F1 of .613, far above every real representation, and yet show no fifths

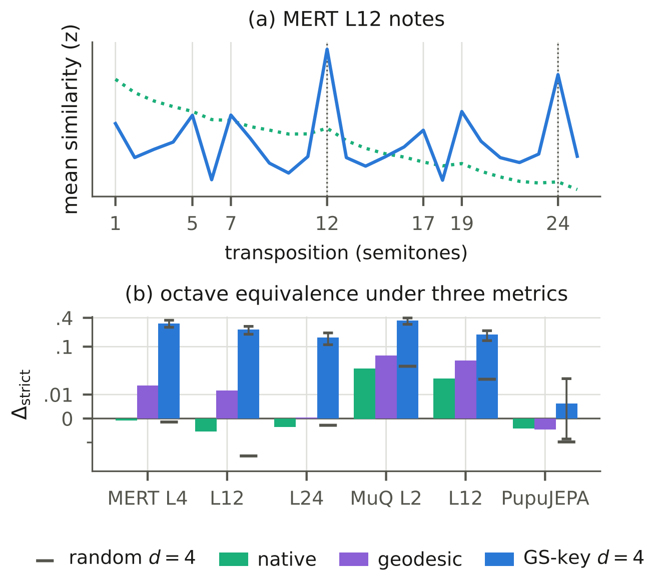

Fig. 2. Two views of the same rank-4 projection, fitted only on GiantSteps track tonics and applied unchanged to individual NSynth recordings. (a) Ambient similarity decays smoothly with transposition distance, while the projected curve reaches maxima at exactly twelve and twenty-four semitones. (b) The strict contrast that curve summarizes, with the matched random subspace as a control. In MERT ambient and random are both near zero; in MuQ they are comparable and above zero, so there the learned projection must be read against the random one. Error bars are instrument-bootstrap 95% intervals.

geometry in either their centroids, at −0.008, or their readout weights, at −0.005. This rules out the reading that any sufficiently accurate 24-way classifier acquires fifths-structured weights because neighboring keys are intrinsically confusable.

The log-CQT outperforms MERT layer 12 under track-grouped cross-validation, at a macro F1 of .392 against .371, despite showing no fifths structure in its class centroids, at −0.035 with p of .52, while its probe weights carry that structure strongly, at 0.560. Tonal relations are therefore linearly accessible in a log-CQT without being expressed in its ambient geometry. Randomly shuffled folds substantially inflate performance for several representations, particularly MERT, MuQ and PQ-STFT, and reverse this comparison, motivating the grouped protocol used throughout.

Across depth the two geometries move in opposite directions in all three learned music encoders. Centroid geometry weakens, from 0.445 to 0.294 in MERT, 0.533 to 0.372 in MuQ and 0.663 to 0.548 in MATPAC, while readout geometry strengthens over the same layers. We report this as a correlation rather than a mechanism, since key F1 does not improve along the way.

The reverse profile is common: MuQ layer 2 shows 0.533 against 0.103, MATPAC layer 6 the strongest centroid geometry of any arm at 0.663 with a chromatic correlation of -0.061, and the two 60-bin-per-octave front ends behave the same way, HCQT at 0.554 against 0.134 and PQ-STFT at 0.550 against 0.172, though resolution is confounded with stacking and normalization. PupuJEPA similarly shows weak ambient geometry, at 0.072 with p of .052, but strong readout geometry at 0.570; its low F1 limits direct comparison. The location of tonal structure therefore differs across representations, including among handcrafted front ends.

4.2. Sample-limited masking

Fig. 1(a) varies pooling duration while holding the number of clips per centroid fixed. A single 13 ms MERT frame already yields 59% of the 20 s value, MuQ starts at 66% of its own, and the chromagram is flat to within 3%, with no knee in the sweep. EnCodec is the exception, rising sixfold.

By contrast, Fig. 1(b) varies clips per centroid at full-clip pooling. MERT L12 rises from 0.035 to 0.468, a 13.4-fold change against 1.7-fold across the thousandfold window sweep, and in the single-example regime fifths geometry is indistinguishable from zero for every arm.

This pattern is consistent with a dominant contribution from ε in Eq. (1), as within-key variation across tracks is progressively reduced when more examples contribute to each centroid. Centroid sample size is therefore the dominant source of the apparent clip-level effect.

4.3. Octave equivalence depends on the metric

Octave equivalence depends strongly on the metric. Under the native metric and the sign-consistency criterion registered in advance, MuQ shows a weak but reliable effect at every layer, its strict contrast falling from 0.037 at layer 2 to 0.022 at layer 12 against 0.698 for the chromagram, whereas MERT and the codec largely fail it. Averaging NSynth recordings into pitch centroids recovers no reliable effect in MERT, indicating that the near-zero geometry is not explained by finite-sample variation alone. A k-nearest-neighbor geodesic metric reveals small but reliable effects in MERT layers 4 and 12 and in EnCodec, stable for k from 5 to 50, but not in layer 24. MERT layer 12 changes from -0.003 under the native metric to 0.012 under the geodesic one and 0.223 after a supervised rank-4 projection. MATPAC follows the same ordering across the three metrics as MuQ, at 0.009, 0.051 and 0.114. Octave structure can therefore be absent from global distances, weakly preserved on the manifold, or visible only after supervised reweighting. The geodesic metric is not uniformly more revealing: for the chromagram, which encodes octave identity directly, the contrast falls from 0.698 to 0.173.

The projection is fitted only to GiantSteps tonic labels, with no notes and no octave information, and applied unchanged to NSynth recordings. It raises the strict contrast to 0.304 for MERT layer 4 and 0.346 for MuQ layer 2, and Fig. 2(a) shows the curve those contrasts come from, with maxima at exactly twelve and twenty-four semitones. In MuQ a matched random rank-4 subspace retains about the native effect, at 0.039 for layer 2, but stays far below the learned projection, while in MERT it is near zero. The learned subspaces account for only 0.08 to 0.29% of NSynth variance in MuQ and 0.11 to 0.13% in MERT. PupuJEPA is a contrasting case in which all three are indistinguishable from zero, and its GiantSteps key F1 is the lowest of any arm, so the absence there is a statement about the metrics we tested rather than about the representation.

Key supervision does not guarantee pitch-height invariance, since GiantSteps key labels may correlate with subgenre, tempo or production, though the transfer rules out an explanation resting only on GiantSteps-specific nuisance correlations. Nor is it mechanistic: whether the projection suppresses or amplifies is not identifiable here.

A registered prediction that failed. We predicted that removing the recovered subspace would at least halve clip- level fifths geometry. Against a baseline recomputed on the same held-out clips it barely moves, from 0.127 to 0.119 for MERT layer 12 and from 0.289 to 0.281 for MuQ layer 2, and only the twelve-dimensional chromagram collapses. The subspace is therefore sufficient to expose the relation without exclusively containing it.

5. DISCUSSION

No single measurement characterizes musical structure in a representation. Probe performance tests linear accessibility, the native metric global visibility, geodesic distance local manifold organization, and projection target-informed linear recoverability. When centroid geometry is weak, varying the number of examples distinguishes sample-limited estimation from stable metric effects, so the number of items behind each centroid should be reported as standard, since the same encoder scores 0.035 or 0.468 depending only on that. These measurements do not form a universal hierarchy, as which metric is most informative depends on how the relation is represented.

Limitations. GiantSteps is entirely electronic dance music, where key correlates with subgenre, and the transfer constrains that confound without removing it. NSynth is monophonic, and our subspaces are linear by design.

6. CONCLUSION

Musical structure does not occupy a fixed geometric form across representations. Decodability, ambient geometry, and readout geometry can dissociate substantially: log-CQT and PupuJEPA show weak centroid structure with strong readouts, and MATPAC, HCQT, PQ-STFT and MuQ the reverse. A relation absent from global distances may be organized on the manifold, far stronger under reweighting, or undetected by all three.

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